



Numerische Implementierung der linearen FEM

Aufgabenblatt 4

1 Allgemeine Hinweise

Die Aufgabenblätter sind als Hausaufgaben in Gruppen zu bearbeiten. Die Abgabe erfolgt per E-Mail an: stefan.hildebrand@tu-berlin.de

Halten Sie Ihre Ergebnisse, Antworten und Erläuterungen für die Aufgaben in einem PDF-Dokument fest und reichen Sie dieses zusammen mit dem Quelltext in einem ZIP-Archiv auf ISIS ein.

Geben Sie auf der ersten Seite des PDF-Dokuments sowie im Quelltext die Namen und Matrikelnummern aller beteiligter Gruppenmitglieder an.

In dieser Lehrveranstaltung können Matlab und GNU Octave verwendet werden.

Quelltext ist zu kommentieren, mindestens sind Eingabe, Ausgabe und Zweck von Funktionen stichpunktartig anzugeben. Wo Algorithmen/ Formeln implementiert werden, ist auf die Quelle zu verweisen (z.B. Nummer der Gleichung im Aufgabenblatt und Name des Algorithmus). Bedeutungen von Variablen/ Parametern müssen eindeutig erkennbar sein, entweder durch gleiche Benennung wie im Aufgabenblatt (z.B. `gamma` für γ) oder durch Kommentare.

Am Ende der Aufgabenstellung finden Sie Materialien mit Formeln und Materialien, die Sie für die Bearbeitung der Aufgabe verwenden und als Referenz heranziehen können.

Implementierte Algorithmen/ Verfahren sind zu validieren. Die Methodik dazu können Sie selbst geeignet wählen. Die Ergebnisse der Validierung ist im PDF-Dokument zusammenzufassen und zu veranschaulichen. Dabei festgestellte Abweichungen sollen begründet und ihre Bedeutung für die Anwendbarkeit des gesamten Codes eingeschätzt werden.

Falls sich im Laufe der Bearbeitung inhaltliche oder organisatorische Schwierigkeiten ergeben, die nicht innerhalb der Gruppe gelöst werden können, findet sich vielleicht eine Lösung in einer individuellen Sprechstunde oder auch per E-Mail an: stefan.hildebrand@tu-berlin.de



2 Einleitung

Nachdem Sie sich erfolgreich selbstständig gemacht haben und regelmäßig Aufträge für spektakuläre Werbeaktionen bearbeiten, meldet sich der Hersteller von Eisenbahnen, bei dem Sie sich vor einem halben Jahr beworben hatten, erneut. Die Simulationen, die Sie damals für Ihr Gespräch vorbereitet hatten, waren auf große Begeisterung gestoßen. Um sich für einen größeren Auftrag zu beweisen, werden Sie gebeten, nun Simulationen unter Berücksichtigung von Plastizität und Materialverhärtung zu erstellen.

Zur Demonstration wählen Sie sich wieder eines der folgenden Bauteile (bzw. einen sinnvollen Ausschnitt):

- Bremsscheibe
- Rad
- Schubstange (einer Dampflok)
- Achse
- Blattfeder (einzelnes Blatt)
- Schienenstück (Querschnitt oder Längsabschnitt)
- Zahn eines Zahnrades
- Bügel eines Stromabnehmers
- ...

Bei der Recherche zu den Abmessungen und Materialeigenschaften erinnern Sie sich, dass für die meisten Bauteile als proprietäre Konstruktionen keine exakten Daten verfügbar sind. Daher schätzen Sie diese Werte ingenieurmäßig sinnvoll ab. Anschließend entwerfen Sie ein einfaches, mechanisches Ersatzmodell. Die Belastungen erhöhen Sie so weit, dass plastisches Verhalten erkennbar wird. Dieses berechnen Sie nun mit Ihrem FE-Code, dokumentieren akkurat das Modell und die Ergebnisse und bewerten die Ergebnisse kritisch. Dazu gehört auch eine Konvergenzbetrachtung, d.h. keine vollständige Konvergenzanalyse, aber eine fundierte Abschätzung der Ergebnisqualität.

Erneut entschließen Sie sich, als konsistentes Einheitensystem (KES) ausschließlich SI-Einheiten zu verwenden.

Viel Erfolg!



3 Residuale, zeitabhängige Form

Sie beginnen mit dem FE-Code für Scheiben (2D, in der Ebene belastet), die statisch belastet sind und rein linear-elastisch reagieren, den Sie damals erstellt haben.¹

Dieser muss so angepasst werden, dass nichtlineare Materialmodelle überhaupt verarbeitet werden können. Statt einer konstanten Steifigkeitsmatrix ist jetzt die Jacobi-Matrix auf Grundlage des jeweiligen algorithmischen Tangenten-Moduls zu verwenden.

Für die Lösung des nichtlinearen Problems ist der Newton-Raphson-Algorithmus zu implementieren.

Weiterhin sollen zeitliche veränderliche Belastungen und zeitabhängiges Materialverhalten berechnet werden können. Der Code muss also auch für quasistatische Anwendungen erweitert werden.

4 Plastisches Materialmodell

Implementieren Sie das Materialmodell für Von Mises-Plastizität mit isotroper und kinematischer Verhärtung für kleine Dehnungen.

Testen Sie dieses Materialmodell für 25CrMo4 und eine kleine Anzahl FE-Elemente (1-4 Stück). Geben Sie ein-achsige zyklische Belastungen (2-4 Zyklen) auf, die zu Dehnungsamplituden von ca. 5% führen (wie in den Materialien beschrieben). Erproben Sie dabei unterschiedliche Verhärtungs-Einstellungen – rein isotrope, rein kinematische, gemischte Verhärtung ($r \in [0, 1]$).

Als Materialparameter können Sie annehmen:

- E-Modul: $E = 205 \text{ GPa}$
- Poisson-Zahl: $\nu = 0.29$
- Initiale Fließgrenze: $\sigma_y^0 = 695 \text{ MPa}$
- Verhärtungsmodul: $H = 2091 \text{ MPa}$
- Kompressionsmodul: $K = \frac{E}{3(1-2\nu)}$
- $\mu = G$ Zweiter Lamé-Parameter, Schubmodul

¹Bzw. mit dem auf ISIS vorgegebenen Code



5 Anwendung

Wählen Sie ein sinnvolles Bauteil, das Sie sinnvoll belasten, bis in den plastischen Bereich hinein. Geben Sie mehrere Belastungszyklen auf – wobei die Belastungen nicht notwendigerweise einen Vorzeichenwechsel erfahren müssen.

Beachten Sie, dass aufgrund des numerischen Aufwands ggf. nur eine relativ grobe Vernetzung berechnet werden kann.

Werten Sie die Ergebnisse wie gewohnt kritisch und anwendungsbezogen aus.



Material

Von Mises plasticity

with isotropic and kinematic hardening for small strains

6 Theory

Remark: the tensor $\mathbf{\kappa}$ represents back stresses for a generic hardening model, including isotropic and kinematic hardening portions. For our case, it will later be split up in a scalar term κ for isotropic hardening and a tensor term $\boldsymbol{\alpha}$ for kinematic hardening.

6.1 Structure of algorithmic material models with internal variables

In the framework of continuum mechanics the concept of internal variables is employed to describe inelastic deformation processes. The assumption of local material behaviour leads to the continuous stress formulation

$$\boldsymbol{\sigma} = \hat{\boldsymbol{\sigma}}(\boldsymbol{\varepsilon}, \mathbf{k}), \quad (6.1)$$

where $\hat{\boldsymbol{\sigma}}$ is a continuous stress function, the arguments of which are the strain tensor $\boldsymbol{\varepsilon}$ as well as a set of internal variables $\mathbf{k}$ which characterise the dissipative properties of a particular material model. The evolution in time of the internal variables $\mathbf{k}$ is given by the evolution equation

$$\dot{\mathbf{k}} = \boldsymbol{\nu}(\boldsymbol{\varepsilon}, \mathbf{k}) \quad \text{with} \quad \mathbf{k}(t = t_0) = \mathbf{k}_0, \quad (6.2)$$

where $\boldsymbol{\nu}$ is a continuous evolution operator and $\mathbf{k}_0$ is an initial value for time step t_0 . This class of inelastic material models is represented by a system of coupled first-order differential equations. Such problems can be solved using a framework of numerical integration in which a particular time interval $[t_n, t_{n+1}]$ is considered, knowing all variables at time step t_n . Classical stress update algorithms compute the stress $\boldsymbol{\sigma}_{n+1}$ at time step t_{n+1} with a preset deformation $\boldsymbol{\varepsilon}_{n+1}$ at time step t_{n+1} and are referred to as *deformation driven algorithmic material models*. A decisive factor for the algorithmic evaluation of the model is the choice of the numerical integration scheme to be used for the set of evolution equations presented in (6.2). The integration algorithm

$$\mathbf{k}_{n+1} = \mathbf{k}_n + \Delta t [(1 - \alpha) \boldsymbol{\nu}(\boldsymbol{\varepsilon}_n, \mathbf{k}_n) + \alpha \boldsymbol{\nu}(\boldsymbol{\varepsilon}_{n+1}, \mathbf{k}_{n+1})] \quad (6.3)$$

represents a suitable form of the stress update algorithm. Here, $\Delta t = t_{n+1} - t_n$ is the time increment and $\alpha \in [0, 1]$ is an integration operator which provides the explicit Euler method for $\alpha = 0$ and relates to the implicit Euler method for $\alpha = 1$. For $\alpha = 0.5$ the trapezoidal rule is obtained which is a second-order method unlike the Euler methods which are first-order methods. Reformulation of equation (6.2) as a deformation driven update algorithm $\hat{\mathbf{k}}_{n+1}$ leads to the symbolic expression

$$\mathbf{k}_{n+1} = \hat{\mathbf{k}}_{n+1}(\boldsymbol{\varepsilon}_{n+1}; \boldsymbol{\varepsilon}_n, \mathbf{k}_n) \quad (6.4)$$



which calculates the algorithmic internal variables $\mathbf{k}_{n+1}$ at time step t_{n+1} based on the preset deformation $\boldsymbol{\varepsilon}_{n+1}$ at time step t_{n+1} and the known variables $\boldsymbol{\varepsilon}_n$ and $\mathbf{k}_n$ at the previous time step t_n . For linear evolution equations the update $\mathbf{k}_{n+1}$ follows directly from equation (6.4) whereas, in the case of non-linear evolution operators $\hat{\mathbf{k}}_{n+1}$, equation (6.4) is an algorithmic box whereby $\mathbf{k}_{n+1}$ is determined iteratively. Now we carry out the algorithmic formulation of equation (6.1)

$$\boldsymbol{\sigma}_{n+1} = \hat{\boldsymbol{\sigma}}(\boldsymbol{\varepsilon}_{n+1}, \mathbf{k}_{n+1}) \quad (6.5)$$

and insert the expression (6.4)

$$\boldsymbol{\sigma}_{n+1} = \hat{\boldsymbol{\sigma}}(\boldsymbol{\varepsilon}_{n+1}, \hat{\mathbf{k}}_{n+1}(\boldsymbol{\varepsilon}_{n+1}; \boldsymbol{\varepsilon}_n, \mathbf{k}_n)). \quad (6.6)$$

The algorithmic stress now can be transformed to the function

$$\boldsymbol{\sigma}_{n+1} = \hat{\boldsymbol{\sigma}}(\boldsymbol{\varepsilon}_{n+1}, \boldsymbol{\varepsilon}_n, \mathbf{k}_n). \quad (6.7)$$

The algorithmic stresses $\boldsymbol{\sigma}_{n+1}$ at the end of time step t_{n+1} are calculated by the preset deformation $\boldsymbol{\varepsilon}_{n+1}$ at time step t_{n+1} and by the history variables $\boldsymbol{\varepsilon}_n$ and $\mathbf{k}_n$ at the beginning of the time interval. It is important that, in the framework of a deformation driven update algorithm, the algorithmic constitutive equations of inelasticity are formally considered to be non-linear depending on the total deformation $\boldsymbol{\varepsilon}_{n+1}$ while the history variables $\boldsymbol{\varepsilon}_n$ and $\mathbf{k}_n$ are considered to be „frozen“.

For the iterative solution of boundary value problems by means of the Finite Element Method and, respectively, the constitutive driver algorithm developed within tutorials 3 and 4, the sensitivities of the algorithmic stress $d\boldsymbol{\sigma}_{n+1}$ with respect to changes of the total deformation $d\boldsymbol{\varepsilon}_{n+1}$ are needed. By means of algorithmic consistent tangent operators $\mathbf{E}_{a,n+1}$, the sensitivities result in

$$d\boldsymbol{\sigma}_{n+1} = \mathbf{E}_{a,n+1} : d\boldsymbol{\varepsilon}_{n+1}. \quad (6.8)$$

Formally they are calculated as the derivatives of the algorithmic stress $\boldsymbol{\sigma}_{n+1}$ with respect to the total deformation $\boldsymbol{\varepsilon}_{n+1}$ at time step t_{n+1} . The tangent moduli

$$\mathbf{E}_{a,n+1} \doteq \frac{d\boldsymbol{\sigma}_{n+1}}{d\boldsymbol{\varepsilon}_{n+1}} \quad (6.9)$$

at time step t_{n+1} are fourth-order tensors which are generally provided in an analytical manner by the constitutive routine. However, by using calculations of perturbations, these can be approximated by so-called „numerical tangent moduli“.

The driver algorithm dealt with in earlier exercises is now to be extended by a constitutive law with internal variables. The underlying structure of the algorithm preserves as it is shown in box 6.1. At each iteration step the constitutive box 6.1 is evaluated for a retained axial deformation $\boldsymbol{\varepsilon}_{n+1}$. The iteration is proceeded as long as the convergence criterion

$$\|\bar{\boldsymbol{\sigma}}\| < tol \quad (6.10)$$

is met, with $tol \ll 1$. After the convergence criterion is met, the internal variables are updated $\mathbf{k}_n \leftarrow \mathbf{k}_{n+1}$ and the time step is increased $t_{n+1} \leftarrow t_{n+1} + \Delta t$ with a new preset axial deformation $\boldsymbol{\varepsilon}_{n+1}(t_{n+1})$.



6.2 Basic equations of classical rate-independent plasticity

1. Kinematics: Additive decomposition of the strain tensor

In the framework of plasticity for small strains, one assumes that the strain tensor ε can be decomposed into an elastic part ε^e and a plastic part ε^p

$$\varepsilon = \varepsilon^e + \varepsilon^p. \quad (6.11)$$

Herein, the plastic strains ε^p are *internal variables*, the time evolution of which is given by the flow rule which is an evolution equation given below, see 4.

2. Elastic stress-strain-relation

The stress tensor σ is related to the elastic strains $\varepsilon^e = \varepsilon - \varepsilon^p$ by means of the strain-energy function Ψ via the hyperelastic relationship

$$\sigma = \frac{\partial \Psi(\varepsilon^e)}{\partial \varepsilon^e}. \quad (6.12)$$

In the framework of linear elasticity, the strain-energy function is expressed by a quadratic function in dependence of ε , i.e. $\Psi = 1/2 \varepsilon^e : \mathbf{E}^e : \varepsilon^e$, where $\mathbf{E}^e$ is the constant, fourth-order elastic tangent moduli tensor. Then, equations (6.11) and (6.12) result in

$$\sigma = \mathbf{E}^e : [\varepsilon - \varepsilon^p]. \quad (6.13)$$

3. Elastic domain in stress space and yield condition

The so-called *yield function* $\Phi(\sigma, \kappa)$ limits the admissible states of $\{\sigma, \kappa\}$ in stress space to a domain $\mathbb{E}$. Mathematically, this is expressed by

$$\mathbb{E} = \{(\sigma, \kappa) | \Phi(\sigma, \kappa) \leq 0\}. \quad (6.14)$$

The interior of $\mathbb{E}$ is described by $\Phi(\sigma, \kappa) < 0$ and is also referred to as *elastic region*, whereas $\Phi(\sigma, \kappa) = 0$ characterizes the boundary of this domain. That boundary is referred to as *yield surface* which represents plastic states. As a remark, all states $\{\sigma, \kappa\}$ outside $\mathbb{E}$ are non-admissible and therefore excluded within the framework of classical rate-independent plasticity.

4. Flow rule and hardening law. Loading/unloading conditions

The irreversibility of plastic flow is represented by the following, not necessarily associative, evolution equations for (ε^p, κ)

$$\dot{\varepsilon}^p = \lambda \nu(\sigma, \kappa) \quad (6.15)$$

$$\dot{\kappa} = \lambda \zeta(\sigma, \kappa), \quad (6.16)$$

where equation (6.15) is denoted as *flow rule* and equation (6.16) is referred to as *hardening law*. Here, ν and ζ are prescribed functions which define the direction of plastic flow as well as the type of hardening.

Within the framework of plasticity, it is common to choose a so-called „associated“ structure, i.e. the functions ν and ζ are derivatives of the *yield function* $\Phi(\sigma, \kappa)$ with respect to σ and κ , as a result of the postulate of maximum plastic dissipation, cf. also the dissipation inequality $\mathcal{D} = \sigma : \dot{\epsilon}^p + \kappa \bullet \dot{k} \geq 0$.

The special associated structure of the flow rule and the hardening law result in

$$\dot{\epsilon}^p = \lambda \frac{\partial \Phi}{\partial \sigma} = \lambda \nu \quad (6.17)$$

$$\dot{k} = -\lambda \frac{\partial \Phi}{\partial \kappa} = -\lambda \zeta. \quad (6.18)$$

The consistency parameter, or, Lagrange multiplier $\lambda \geq 0$ has to satisfy the so-called *Kuhn-Tucker complementary conditions*

$$\lambda \geq 0, \quad \Phi(\sigma, \kappa) \leq 0, \quad \lambda \Phi(\sigma, \kappa) = 0, \quad (6.19)$$

which are also referred to as *loading/unloading conditions*. In addition, λ has to satisfy the *consistency condition*

$$\lambda \dot{\Phi}(\sigma, \kappa) = 0. \quad (6.20)$$

The Kuhn-Tucker complementary conditions can be explained and interpreted according to the following settings:

- a) A state $\{\sigma, \kappa\}$ *inside* of $\mathbb{E}$ is considered such that $\Phi(\sigma, \kappa) < 0$ holds. From condition (6.19)₃ we conclude that

$$\lambda \Phi = 0 \quad \text{and} \quad \Phi < 0 \quad \longrightarrow \quad \lambda = 0. \quad (6.21)$$

From equation (6.15) we observe that the internal variables are not evolving within the considered time interval, such that the material behaviour is *purely elastic* and $\dot{\epsilon}^p = \mathbf{0}$ as well as $\dot{k} = \mathbf{0}$.

- b) A state $\{\sigma, \kappa\}$ *on the boundary* of $\mathbb{E}$ is considered such that $\Phi(\sigma, \kappa) = 0$ holds. In this case, condition (6.19)₃ is fulfilled automatically, even if $\lambda > 0$. Whether λ is positive, negative or zero, can be decided by means of condition (6.20). Hereby, two different cases are to be differed:

- i. If $\dot{\Phi}(\sigma, \kappa) < 0$, from condition (6.20) we obtain

$$\lambda \dot{\Phi} = 0 \quad \text{and} \quad \dot{\Phi} < 0 \quad \longrightarrow \quad \lambda = 0. \quad (6.22)$$

From equation (6.15), in turn, it follows that the internal variables do not evolve over time, i.e. $\dot{\epsilon}^p = \mathbf{0}$ and $\dot{k} = \mathbf{0}$. Because $\{\sigma, \kappa\}$ lies *on the boundary* of $\mathbb{E}$, this state is called *elastic unloading*.

- ii. If $\dot{\Phi}(\sigma, \kappa) = 0$, condition (6.20) is fulfilled automatically. If $\lambda > 0$, $\dot{\epsilon}^p \neq \mathbf{0}$ and $\dot{k} \neq \mathbf{0}$ hold. This state is called *plastic loading*. In the case of $\lambda = 0$ one speaks of *neutral loading*.

6.3 Von Mises plasticity with isotropic and kinematic hardening

Within the framework of metal plasticity, the internal plastic variables are typically given as $\mathbf{k} = \{k, \mathbf{a}\}$. Here, k indicates the equivalent plastic strain that defines the isotropic hardening of the von Mises yield surface, and $\mathbf{a}$ defines the center of the von Mises yield surface in stress deviator space at kinematic hardening. By application of the quadratic form of the free energy

$$\Psi = \frac{K}{2} [\boldsymbol{\varepsilon}^e : \mathbf{I}]^2 + G [\boldsymbol{\varepsilon}_{\text{dev}}^e]^2 : \mathbf{I} + \frac{H}{2} r k^2 + \frac{H}{2} [1 - r] a_e^2 \quad (6.23)$$

with

$$a_e = \sqrt{\frac{2}{3}} \|\mathbf{a}\|, \quad (6.24)$$

the elastic stresses can be expressed by

$$\boldsymbol{\sigma} = \frac{\partial \Psi}{\partial \boldsymbol{\varepsilon}^e} = \boldsymbol{\sigma}_{\text{vol}} + \boldsymbol{\sigma}_{\text{dev}} = \sigma_m \mathbf{I} + \boldsymbol{\sigma}_{\text{dev}} \quad (6.25)$$

with the volumetric and deviatoric parts

$$\sigma_m = K \boldsymbol{\varepsilon}_{\text{vol}}^e = K [\boldsymbol{\varepsilon}^e : \mathbf{I}], \quad \text{respectively} \quad \boldsymbol{\sigma}_{\text{dev}} = 2 G \boldsymbol{\varepsilon}_{\text{dev}}^e = 2 G [\boldsymbol{\varepsilon}_{\text{dev}} - \boldsymbol{\varepsilon}^p] \quad (6.26)$$

as well as the hardening stresses

$$\kappa = -\frac{\partial \Psi}{\partial k} = -r H k \quad (6.27)$$

$$\boldsymbol{\alpha} = -\frac{\partial \Psi}{\partial \mathbf{a}} = -\frac{2}{3} [1 - r] H \mathbf{a}. \quad (6.28)$$

Furthermore, combining the von Mises yield function

$$\Phi = \|\boldsymbol{\sigma}_{\text{dev}}^{\text{red}}\| - \sqrt{\frac{2}{3}} [\sigma_y - \kappa] \leq 0 \quad (6.29)$$

which includes the reduced deviatoric stress

$$\boldsymbol{\sigma}_{\text{dev}}^{\text{red}} = \boldsymbol{\sigma}_{\text{dev}} + \boldsymbol{\alpha}, \quad (6.30)$$

with the hardening laws (6.18), leads to the associated evolution equations

$$\dot{\boldsymbol{\varepsilon}}^p = \lambda \boldsymbol{\nu}, \quad \boldsymbol{\nu} = \frac{\partial \Phi}{\partial \boldsymbol{\sigma}} = \frac{\boldsymbol{\sigma}_{\text{dev}}^{\text{red}}}{\|\boldsymbol{\sigma}_{\text{dev}}^{\text{red}}\|} \quad (6.31)$$

$$\dot{k} = \lambda \zeta_\kappa, \quad \zeta_\kappa = \frac{\partial \Phi}{\partial \kappa} = \sqrt{\frac{2}{3}} \quad (6.32)$$

$$\dot{\mathbf{a}} = \lambda \zeta_\alpha, \quad \zeta_\alpha = \frac{\partial \Phi}{\partial \mathbf{a}} = \frac{\boldsymbol{\sigma}_{\text{dev}}^{\text{red}}}{\|\boldsymbol{\sigma}_{\text{dev}}^{\text{red}}\|} = \boldsymbol{\nu}. \quad (6.33)$$

Here, H represents the so-called hardening modulus, σ_y represents the yield stress, $r \in [0, 1]$ is a parameter which specifies the hardening mechanism, i.e. $r = 1$: isotropic hardening $r = 0$: kinematic hardening, and $\boldsymbol{\nu}$ represents the normalised tensor orthogonal to the von Mises yield surface.

6.4 Algorithmic structure

The algorithm developed within this section applies to three-dimensional stress states as well as to plane-strain cases. The simplicity of the von Mises yield criterion (6.29) enables a closed-form solution without the need of additional local Newton iterations. This algorithm is called „radial return algorithm“.

Radial return mapping

Within the previous section, the internal plastic variables were introduced as $\mathbf{k} = \{k, \mathbf{a}\}$. Furthermore, the reduced deviatoric stress was defined as

$$\boldsymbol{\sigma}_{\text{dev}}^{\text{red}} = \boldsymbol{\sigma}_{\text{dev}} + \boldsymbol{\alpha}. \quad (6.34)$$

At the the end of a time step $[t_n, t_{n+1}]$, the normalised tensor, orthogonal to the von Mises yield surface, reads

$$\boldsymbol{\nu}_{n+1} = \frac{\boldsymbol{\sigma}_{\text{dev } n+1}^{\text{red tr}}}{\|\boldsymbol{\sigma}_{\text{dev } n+1}^{\text{red tr}}\|}. \quad (6.35)$$

By means of the Backward Euler Method, the evolution equations (6.31)-(6.33) transform to the discretised structure

$$\boldsymbol{\varepsilon}_{n+1}^{\text{p}} = \boldsymbol{\varepsilon}_n^{\text{p}} + \Delta\lambda_{n+1} \boldsymbol{\nu}_{n+1} \quad (6.36)$$

$$k_{n+1} = k_n + \Delta\lambda_{n+1} \sqrt{2/3} \quad (6.37)$$

$$\mathbf{a}_{n+1} = \mathbf{a}_n + \Delta\lambda_{n+1} \boldsymbol{\nu}_{n+1} \quad (6.38)$$

where $\Delta\lambda = \Delta t \lambda$.

Deviatoric reduced stresses. The deviatoric reduced stresses $\boldsymbol{\sigma}_{\text{dev } n+1}^{\text{red}}$ specified in equation (6.34) are calculated as

$$\boldsymbol{\sigma}_{\text{dev } n+1}^{\text{red}} = \boldsymbol{\sigma}_{\text{dev } n+1} + \boldsymbol{\alpha}_{n+1} \quad (6.39)$$

$$= 2G [\boldsymbol{\varepsilon}_{\text{dev } n+1} - \boldsymbol{\varepsilon}_{n+1}^{\text{p}}] - \frac{2}{3} [1 - r] H \mathbf{a}_{n+1}, \quad (6.40)$$

by means of equation (6.33). Inserting the evolution equations (6.36) and (6.38) leads to

$$\boldsymbol{\sigma}_{\text{dev } n+1}^{\text{red}} = \boldsymbol{\sigma}_{\text{dev } n+1}^{\text{red tr}} - \Delta\lambda_{n+1} \left[2G + \frac{2}{3} [1 - r] H \right] \boldsymbol{\nu}_{n+1}, \quad (6.41)$$

where

$$\boldsymbol{\sigma}_{\text{dev } n+1}^{\text{red tr}} = \boldsymbol{\sigma}_{\text{dev } n+1}^{\text{tr}} + \boldsymbol{\alpha}_{n+1}^{\text{tr}} \quad (6.42)$$

with

$$\boldsymbol{\sigma}_{\text{dev } n+1}^{\text{tr}} = 2G [\boldsymbol{\varepsilon}_{\text{dev } n+1} - \boldsymbol{\varepsilon}_n^{\text{p}}] \quad (6.43)$$

and

$$\alpha_{n+1}^{\text{tr}} = -\frac{2}{3} [1 - r] H \mathbf{a}_n. \quad (6.44)$$

Consistency parameter. To determine the consistency parameter $\Delta\lambda_{n+1}$, we recall that by definition $\sigma_{\text{dev } n+1}^{\text{red}} = \|\sigma_{\text{dev } n+1}^{\text{red}}\| \nu_{n+1}$ holds. Hence, the normalised tensor ν_{n+1} is solely calculated by means of the elastic trial-stresses $\sigma_{\text{dev } n+1}^{\text{red tr}}$

$$\nu_{n+1} = \frac{\sigma_{\text{dev } n+1}^{\text{red tr}}}{\|\sigma_{\text{dev } n+1}^{\text{red tr}}\|}. \quad (6.45)$$

The von Mises yield criterion according to equation (6.29) reads

$$\Phi_{n+1} = \|\sigma_{\text{dev } n+1}^{\text{red}}\| - \sqrt{\frac{2}{3}} [\sigma_y - \kappa_{n+1}]. \quad (6.46)$$

By means of the hardening law (6.27) and by the evolution equation (6.37) as well as with the definition

$$\Phi_{n+1}^{\text{tr}} = \|\sigma_{\text{dev } n+1}^{\text{red tr}}\| - \sqrt{\frac{2}{3}} [\sigma_y - \kappa_n^{\text{tr}}] \quad \text{with} \quad \kappa_n^{\text{tr}} = -r H k_n, \quad (6.47)$$

the consistency parameter is calculated by

$$\Delta\lambda_{n+1} = \frac{\Phi_{n+1}^{\text{tr}}}{2G + \frac{2}{3} H}. \quad (6.48)$$

The next step is the determination of the consistent elasto-plastic tangent moduli by exact linearisation of the return mapping algorithm. These moduli link incremental strains and incremental stresses and are decisive for the solution of general boundary value problems.

Exact linearisation of the algorithm

Differentiation of the algorithmic expression of the stress tensor

$$\sigma_{n+1} = \sigma_{\text{vol } n+1} + \sigma_{\text{dev } n+1} \quad (6.49)$$

leads to

$$d\sigma_{n+1} = \mathbf{E}_{a n+1} : d\epsilon_{n+1} \quad (6.50)$$

with the elasto-plastic tangent moduli

$$\mathbf{E}_{a n+1} := \frac{d\sigma(\epsilon_{n+1})}{d\epsilon_{n+1}}, \quad (6.51)$$

which is specified below.

By partial differentiation of the volumetric stresses

$$\sigma_{\text{vol } n+1} = K [\epsilon_{n+1} : \mathbf{I}] \mathbf{I} \quad (6.52)$$

as well as the deviatoric stresses

$$\boldsymbol{\sigma}_{\text{dev } n+1} = \boldsymbol{\sigma}_{\text{dev } n+1}^{\text{tr}} - \Delta\lambda_{n+1} \left[2G + \frac{2}{3}[1-r]H \right] \boldsymbol{\nu}_{n+1} \quad (6.53)$$

we obtain

$$\mathbf{E}_{a \, n+1} = \frac{d\boldsymbol{\sigma}_{n+1}}{d\boldsymbol{\varepsilon}_{n+1}} = \frac{d\boldsymbol{\sigma}_{n+1}^{\text{vol}}}{d\boldsymbol{\varepsilon}_{n+1}} + \frac{d\boldsymbol{\sigma}_{n+1}^{\text{dev}}}{d\boldsymbol{\varepsilon}_{n+1}} \quad (6.54)$$

with the volumetric part

$$\frac{d\boldsymbol{\sigma}_{n+1}^{\text{vol}}}{d\boldsymbol{\varepsilon}_{n+1}} = \frac{d}{d\boldsymbol{\varepsilon}_{n+1}} [K [\boldsymbol{\varepsilon}_{n+1} : \mathbf{I}] \mathbf{I}] = K \mathbf{I} \otimes \mathbf{I} \quad (6.55)$$

and the deviatoric part

$$\frac{d\boldsymbol{\sigma}_{n+1}^{\text{dev}}}{d\boldsymbol{\varepsilon}_{n+1}} = \frac{d}{d\boldsymbol{\varepsilon}_{n+1}} \left[\boldsymbol{\sigma}_{\text{dev } n+1}^{\text{tr}} - \Delta\lambda_{n+1} \left[2G + \frac{2}{3}[1-r]H \right] \boldsymbol{\nu}_{n+1} \right] \quad (6.56)$$

$$= \frac{\partial \boldsymbol{\sigma}_{\text{dev } n+1}^{\text{tr}}}{\partial \boldsymbol{\varepsilon}_{n+1}} - \left[2G + \frac{2}{3}[1-r]H \right] \left[\frac{\partial \Delta\lambda_{n+1}}{\partial \boldsymbol{\varepsilon}_{n+1}} \boldsymbol{\nu}_{n+1} + \Delta\lambda_{n+1} \frac{\partial \boldsymbol{\nu}_{n+1}}{\partial \boldsymbol{\varepsilon}_{n+1}} \right]. \quad (6.57)$$

Here, the partial derivatives of the deviatoric trial-stresses

$$\frac{\partial \boldsymbol{\sigma}_{\text{dev } n+1}^{\text{tr}}}{\partial \boldsymbol{\varepsilon}_{n+1}} = \frac{\partial}{\partial \boldsymbol{\varepsilon}_{n+1}} [2G[\boldsymbol{\varepsilon}_{n+1}^{\text{dev}} - \boldsymbol{\varepsilon}_n^{\text{p}}]] = \frac{\partial}{\partial \boldsymbol{\varepsilon}_{n+1}} [2G [\mathbf{I}_{\text{dev}}^{\text{sym}} : \boldsymbol{\varepsilon}_{n+1} - \boldsymbol{\varepsilon}_n^{\text{p}}]] = 2G \mathbf{I}_{\text{dev}}^{\text{sym}}, \quad (6.58)$$

the partial derivatives of the Lagrange multipliers

$$\frac{\partial \Delta\lambda_{n+1}}{\partial \boldsymbol{\varepsilon}_{n+1}} = \frac{\partial}{\partial \boldsymbol{\varepsilon}_{n+1}} \left[\frac{\Phi_{n+1}^{\text{tr}}}{2G + \frac{2}{3}H} \right] = \frac{1}{2G + \frac{2}{3}H} \underbrace{\left[\frac{\partial \Phi_{n+1}^{\text{tr}}}{\partial \boldsymbol{\sigma}_{\text{dev } n+1}^{\text{red tr}}} : \frac{\boldsymbol{\sigma}_{\text{dev } n+1}^{\text{red tr}}}{\partial \boldsymbol{\varepsilon}_{n+1}} \right]}_{= \boldsymbol{\nu}_{n+1} : 2G \mathbf{I}_{\text{dev}}^{\text{sym}} = 2G \boldsymbol{\nu}_{n+1}} = \frac{2G}{2G + \frac{2}{3}H} \boldsymbol{\nu}_{n+1} \quad (6.59)$$

as well as the yield direction

$$\begin{aligned} \frac{\partial \boldsymbol{\nu}_{n+1}}{\partial \boldsymbol{\varepsilon}_{n+1}} &= \frac{\partial}{\partial \boldsymbol{\varepsilon}_{n+1}} \left[\frac{\boldsymbol{\sigma}_{\text{dev } n+1}^{\text{red tr}}}{\|\boldsymbol{\sigma}_{\text{dev } n+1}^{\text{red tr}}\|} \right] = \frac{\partial \|\boldsymbol{\sigma}_{\text{dev } n+1}^{\text{red tr}}\|^{-1}}{\partial \boldsymbol{\varepsilon}_{n+1}} \otimes \boldsymbol{\sigma}_{\text{dev } n+1}^{\text{red tr}} + \|\boldsymbol{\sigma}_{\text{dev } n+1}^{\text{red tr}}\|^{-1} \frac{\partial \boldsymbol{\sigma}_{\text{dev } n+1}^{\text{red tr}}}{\partial \boldsymbol{\varepsilon}_{n+1}} \\ &= \frac{\partial \|\boldsymbol{\sigma}_{\text{dev } n+1}^{\text{red tr}}\|^{-1}}{\partial \boldsymbol{\sigma}_{\text{dev } n+1}^{\text{red tr}}} : \frac{\partial \boldsymbol{\sigma}_{\text{dev } n+1}^{\text{red tr}}}{\partial \boldsymbol{\varepsilon}_{n+1}} \otimes \boldsymbol{\sigma}_{\text{dev } n+1}^{\text{red tr}} + \frac{2G}{\|\boldsymbol{\sigma}_{\text{dev } n+1}^{\text{red tr}}\|} \mathbf{I}_{\text{dev}}^{\text{sym}} \\ &= \frac{2G}{\|\boldsymbol{\sigma}_{\text{dev } n+1}^{\text{red tr}}\|} [\mathbf{I}_{\text{dev}}^{\text{sym}} - \boldsymbol{\nu}_{n+1} \otimes \boldsymbol{\nu}_{n+1}] \end{aligned} \quad (6.60)$$

are calculated as specified above. Combining the single expressions leads to

$$\mathbf{E}_{a \, n+1} = K \mathbf{I} \otimes \mathbf{I} + c_1 \mathbf{I}_{\text{dev}}^{\text{sym}} + c_2 \boldsymbol{\nu}_{n+1} \otimes \boldsymbol{\nu}_{n+1} \quad (6.61)$$

with

$$c_1 = 2G \left[1 - \frac{2G + \frac{2}{3}[1-r]H}{\|\boldsymbol{\sigma}_{\text{dev } n+1}^{\text{red tr}}\|} \Delta\lambda_{n+1} \right] \quad (6.62)$$

$$c_2 = 2G \left[2G + \frac{2}{3}[1-r]H \right] \left[\frac{\Delta\lambda_{n+1}}{\|\boldsymbol{\sigma}_{\text{dev } n+1}^{\text{red tr}}\|} - \frac{1}{2G + \frac{2}{3}H} \right] \quad (6.63)$$

6.5 Radial Return Algorithm: Von Mises plasticity with associated isotropic and kinematic hardening

a) Given: Strains and internal variables of the previous state of equilibrium

$$\varepsilon_{n+1}, \{\varepsilon_n^p, k_n, \mathbf{a}_n\}$$

b) Computation of the trial values

$$\begin{aligned}\sigma_{\text{vol } n+1} &= K [\varepsilon_{n+1} : \mathbf{I}] \mathbf{I} \\ \sigma_{\text{dev } n+1}^{\text{tr}} &= 2 G [\varepsilon_{n+1}^{\text{dev}} - \varepsilon_n^p] \\ \kappa_{n+1}^{\text{tr}} &= -r H k_n \\ \alpha_{n+1}^{\text{tr}} &= -\frac{2}{3} [1-r] H \mathbf{a}_n \\ \sigma_{\text{dev } n+1}^{\text{red tr}} &= \sigma_{\text{dev } n+1}^{\text{tr}} + \alpha_{n+1}^{\text{tr}} \\ \Phi_{n+1}^{\text{tr}} &= \|\sigma_{\text{dev } n+1}^{\text{red tr}}\| - \sqrt{\frac{2}{3}} [\sigma_y - \kappa_{n+1}^{\text{tr}}]\end{aligned}$$

c) Check yield condition

if $\Phi_{n+1}^{\text{tr}} \leq 0$ set $[\bullet] = [\bullet]_{n+1}^{\text{tr}}$ and go to e) with $c_1 = 2 G$ and $c_2 = 0$, else go to d).

d) Radial return

$$\begin{aligned}\Delta \lambda_{n+1} &= \frac{\Phi_{n+1}^{\text{tr}}}{2 G + \frac{2}{3} H} \\ k_{n+1} &= k_n + \Delta \lambda_{n+1} \sqrt{\frac{2}{3}} \\ \nu_{n+1} &= \frac{\sigma_{\text{dev } n+1}^{\text{red tr}}}{\|\sigma_{\text{dev } n+1}^{\text{red tr}}\|} \\ \mathbf{a}_{n+1} &= \mathbf{a}_n + \Delta \lambda_{n+1} \nu_{n+1} \\ \varepsilon_{n+1}^p &= \varepsilon_n^p + \Delta \lambda_{n+1} \nu_{n+1} \\ \sigma_{\text{dev } n+1} &= \sigma_{\text{dev } n+1}^{\text{tr}} - \Delta \lambda_{n+1} \left[2 G + \frac{2}{3} [1-r] H \right] \nu_{n+1} \\ \sigma_{n+1} &= \sigma_{\text{vol } n+1} + \sigma_{\text{dev } n+1} \\ \kappa_{n+1} &= -r H k_{n+1} \\ \alpha_{n+1} &= -\frac{2}{3} [1-r] H \mathbf{a}_{n+1}\end{aligned}$$

e) Consistent elasto-plastic tangent moduli

$$\mathbf{E}_{a, n+1} = K \mathbf{I} \otimes \mathbf{I} + c_1 \mathbf{I}_{\text{dev}}^{\text{sym}} + c_2 \nu_{n+1} \otimes \nu_{n+1}$$

with

$$\begin{aligned}c_1 &= 2 G \left[1 - \frac{2 G + \frac{2}{3} [1-r] H}{\|\sigma_{\text{dev } n+1}^{\text{red tr}}\|} \Delta \lambda_{n+1} \right] \\ c_2 &= 2 G \left[2 G + \frac{2}{3} [1-r] H \right] \left[\frac{\Delta \lambda_{n+1}}{\|\sigma_{\text{dev } n+1}^{\text{red tr}}\|} - \frac{1}{2 G + \frac{2}{3} H} \right]\end{aligned}$$

Given as **I4dev**
in the template source code